

Factors to consider for Estimators, CIs, and HTs	Population Correlation
Sample Correlation	Properties of Sample Correlation
Simple Linear Regression (SLR) Model	Four Assumptions about data for SLR Model
Distribution of Y_i in SLR	MLEs of β_0 and β_1
Fitted Values	Residuals

A measure of the strength of the linear relationship between two quantitative variables in a population of interest	<table><tr><th>Method</th><th>Consideration 1</th><th>Consideration 2</th></tr><tr><td>Estimator</td><td>Unbiased</td><td>Minimum Variance</td></tr><tr><td>CI</td><td>Proper Coverage</td><td>Minimum Width</td></tr><tr><td>HT</td><td>Proper Size</td><td>Highest Power</td></tr></table>	Method	Consideration 1	Consideration 2	Estimator	Unbiased	Minimum Variance	CI	Proper Coverage	Minimum Width	HT	Proper Size	Highest Power
Method	Consideration 1	Consideration 2											
Estimator	Unbiased	Minimum Variance											
CI	Proper Coverage	Minimum Width											
HT	Proper Size	Highest Power											
• Always between -1 and 1 • Strongly influenced by outliers • It only measures linear association, so the sample correlation could be 0 even in the presence of a strong relationship between x and y	A statistic used to estimate the population correlation												
1. The observations are independent 2. The mean of Y_i is a linear function of x_i 3. The Y_i have a common variance, σ^2, which is the same for every value of x 4. The errors (and hence the Y_i) are normally distributed	$Y_i = \beta_0 + \beta_1 x_i + \epsilon_i \quad \text{for } i = 1, 2, \dots, n$ where ϵ_i is the random error associated with each observation such that each is iid according to a normal distribution such that $\epsilon_i \sim N(0, \sigma^2)$												
• $\hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}$ • $\hat{\beta}_1 = S_{XY}/S_{XX}$	$Y_i \sim N(\beta_0 + \beta_1 x_i, \sigma^2)$												
$e_i = y_i - \hat{y}_i = y_i - \left(\hat{\beta}_0 + \hat{\beta}_1 x_i \right)$	$\hat{y}_i = \hat{\beta}_0 + \hat{\beta}_1 x_i$												

Least Squares Criteria	Estimator for σ^2
Interpretation of Slope Estimate	Interpretation of Intercept Estimate
Sampling Distribution of $\hat{\beta}_1$	Sampling Distribution of $\hat{\beta}_0$
CI's for β_0 and β_1	Sum of Squares Relationship
SST	SSR/SSreg

$\hat{\sigma}^2 = s^2 = \frac{\sum e_i^2}{n-2} = \frac{RSS}{n-2} = \frac{\sum (y_i - \hat{\beta}_0 - \hat{\beta}_1 x_i)^2}{n-2}$	$Q = \sum_{i=1}^n e_i^2 = \sum_{i=1}^n \left(y_i - (\hat{\beta}_0 + \hat{\beta}_1 x_i) \right)^2$
On average, the value of y is $\hat{\beta}_0$ when x is equal to 0	For every one unit increase in x, we get an average increase of $\hat{\beta}_1$ in y
$\hat{\beta}_0 \sim N \left(\beta_0, \sigma^2 \left[\frac{1}{n} + \frac{\bar{x}^2}{S_{XX}} \right] \right)$	$\hat{\beta}_1 \sim N \left(\beta_1, \frac{\sigma^2}{S_{XX}} \right)$
SST = SSreg + RSS = SSR + SSE	• $\hat{\beta}_0 \pm t_{n-2, 1-\alpha/2} \text{se}(\hat{\beta}_0)$ • $\hat{\beta}_1 \pm t_{n-2, 1-\alpha/2} \text{se}(\hat{\beta}_1)$
• SSR = SSreg = $\sum (\hat{y}_i - \bar{y})^2$ • Variability explained by the model • How each fitted value differed from the mean	• SST = $\sum (y_i - \bar{y})^2$ • Total variability in the observed values of y • How each observed value differs from the mean

SSE/RSS	Mean Square
ANOVA table for SLR	Coefficient of Determination
Dummy Variable	Least Absolute Value (LAV) Regression
M-Estimators	Minimizing M-Estimators
Iteratively Reweighted Least Squares (IRLS)	SLR Assumptions

The sum of squares divided by its degrees of freedom	• $SSE = RSS = \sum (y_i - \hat{y}_i)^2$ • Variability NOT explained by the model • How each observed value differs from the fitted line																				
• $R^2 = \frac{SSR}{SST} = 1 - \frac{SSE}{SST}$ • Represents the proportion of variance in y that is accounted for by regression on x. • It is always a value between 0 and 1 • Can be obtained by squared the sample correlation coefficient r	<table><tr><th>Source</th><th>df</th><th>SS</th><th>MS</th><th>F</th></tr><tr><td>Regression</td><td>1</td><td>SSR</td><td>$MSR = \frac{SSR}{1}$</td><td>$\frac{MSR}{MSE}$</td></tr><tr><td>Error</td><td>$n - 2$</td><td>SSE</td><td>$MSE = \frac{SSE}{n - 2}$</td><td></td></tr><tr><td>Total</td><td>$n - 1$</td><td>SST</td><td></td><td></td></tr></table>	Source	df	SS	MS	F	Regression	1	SSR	$MSR = \frac{SSR}{1}$	$\frac{MSR}{MSE}$	Error	$n - 2$	SSE	$MSE = \frac{SSE}{n - 2}$		Total	$n - 1$	SST		
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Estimates of β_0 and β_1 are found by obtaining the values that numerically minimize $\min_{\beta_0, \beta_1} \sum_{i=1} y_i - \hat{y}_i $ • Good at catching outliers in vertical direction • Bad at catching outliers in horizontal direction	• A new variable created from a categorical variable • Will always be one fewer dummy variables than number of categories in categorical variable • Example: Categorical Variable - New or Existing $x = \begin{cases} 1, & \text{if New} \\ 0, & \text{if Existing} \end{cases}$																				
Instead of taking derivative of $\rho(\cdot)$, use a function on the weights that decreases as the size of the residual increases, $\sum_{i=1}^n w(e_i / \hat{\sigma}_e) \mathbf{x}_i = 0$ The issue with minimizing $\sum_{i=1}^n \rho(e_i)$ is that we don't have an initial value of e_i. Iteratively reweighted least squares (IRLS) solves this problem.	Estimates of β_0 and β_1 are the values that minimize $\sum_{i=1}^n \rho(y_i - \hat{y}_i) = \sum_{i=1}^n \rho(e_i)$ where $\rho(\cdot)$ is a function of the residuals that does not increase as rapidly as the square function does.																				
1. The observations are independent 2. The mean of Y_i is a linear function of x_i 3. The Y_i have a common variance σ^2, which is the same for every value of x 4. The errors (and hence the Y_i) are normally distributed	1. Get pilot estimates of β_0 and β_1 (usually with LS estimators). Denote $\hat{\beta}^{(0)} = \left(\hat{\beta}_0^{(0)}, \hat{\beta}_1^{(0)}\right)^{\top}$ 2. Obtain residuals based on $\hat{\beta}^{(0)}$ (Denoted $e_i^{(0)}$) 3. Choose weight function & get preliminary weights (Denoted $w\left(e_i^{(0)}\right), i = 1, \cdots, n$) 4. Estimate β_0 & β_1 by minimizing $Q = \sum w\left(e_i^{(0)}\right)$ 5. Obtain new residuals and new weights 6. Repeat steps 4 and 5 until convergence																				

Checking Independence Assumption	Remedies for Autocorrelation
Checking Linear Relationship Assumption	Outlier
Checking Constant Variance (Homoscedasticity) Assumption	Checking Normally Distributed Errors Assumption
NOTE:	Types of Outliers
Leverage Points	Bad Leverage Point

• For positive serial correlation, you can add lags of the dependent and/or independent variable to the model • For negative serial correlation, check to make sure that none of your variables are overdifferenced • For seasonal correlation, you can add seasonal dummy variables to the model	Use the Durbin-Watson test (test stat ranges from 0 to 4) • $d = 2$ indicates no autocorrelation • $d < 2$ indicates positive serial correlation • $d > 2$ indicates negative serial correlation If d is less than 1.5 or greater than 2.5, there is evidence of autocorrelation
• In small to moderate datasets, an outlier in a linear regression model is flagged as any point whose standardized residual falls outside of the range from -2 to 2. • In large datasets, the accepted range is increased to -4 to 4.	• First plot the data with a scatterplot and check for a linear relationship • Plot the residuals (y) versus the fitted values (x) from the regression model
• Create a normal QQ-plot of the standardized residuals • Conduct the Shapiro-Wilks test for normality on the standardized residuals	Plot the square root of the absolute standardized residuals (y) against the fitted values (x) and check for any nonconstant trend
• Outlier in X Direction: high leverage; low residual • Outlier in Y Direction: low leverage; high residual • Outlier in X and Y Directions: high leverage, high residual	The most critical assumptions are homoscedasticity and independent observations. Without these, the variances derived for $\hat{\beta}_0$ and $\hat{\beta}_1$ do not hold
• A leverage point whose y-value does not follow the pattern set by other data points • An outlier	• Data points that exercise considerable influence on the fitted model. Specifically, they are those points whose x-value is distant from the other x-values. • Leverage points can either be “good” or “bad”

Good Leverage Point	Leverage Rule of Thumb
NOTE:	Cook's Distance
Relation between Cook's Distance and Leverage	Box-Cox Power Transformations
NOTE:	Reasons to use Weighted Least Squares (WLS)
SLR Model via WLS	How do the weights, w_i, affect the variance?

$\sum h_i = k + 1$ Where h_i is a single leverage point and k is the number of explanatory variables. Leverage values greater than $2(k + 1)/n$ or $3(k + 1)/n$ are regarded as high.	• A leverage point whose y-value does follow the pattern set by other data points • Not an outlier
A measure of the influence that the ith observation has on the least squares fit of a regression model, defined by $D_i = \frac{\sum_{j=1}^n (\hat{y}_{j(i)} - \hat{y}_j)^2}{2\hat{\sigma}^2}$ where $\hat{y}_{j(i)}$ denotes the jth fitted value based on the fit obtained when the ith case has been deleted from the fit	Good leverage points still need to be addressed because they inflate R^2
• A systematic method for choosing a power transformation, λ, for a strictly positive response • The set of power transformations is $Y^{(\lambda)} = \begin{cases} \frac{(Y^\lambda - 1)}{\lambda}, & \lambda \neq 0 \\ \log(Y), & \lambda = 0 \end{cases}$	It can be shown that $D_i = \frac{r_i^2}{2} \frac{h_i}{(1 - h_i)}$ so large values of Cook's distance can occur when either the standardized residual for the ith point is large, the leverage for the ith point is large, or both.
1. Stabilize Variance (Dealing with heteroscedasticity) 2. Robust Regression 3. Gives more weight to observations with more information in them	• Regression through the origin by removing the intercept should only be forced if there is a compelling reason to do so, such as data has been observed near the origin • If no compelling reason exists and the line is forced through the origin, it may make the regression worse
• Large w_i implies (x_i, y_i) is important/has a lot of information and we want the fitted line to care a lot about this point • Small w_i implies the opposite: (x_i, y_i) is less important	$Y_i = \beta_0 + \beta_1 x_i + \epsilon_i, \quad \epsilon \stackrel{iid}{\sim} N(0, \sigma^2/w_i)$

Steps for WLS	Choosing Weights
Polynomial Regression	Polynomial Regression Model that solves Overfitting Problem
Interaction Term	Multiple Linear Regression (MLR) Model
Distribution of Y in MLR Model	MLR Model Assumption
MLR Model (Matrix Representation)	Estimate of β

- The choice for weights is not always obvious - An important special case is when Y_i is the average or median of n_i observations. Then the variance of Y_i is taken to be proportional to $1/n_i$, and w_i is taken to be n_i.	- Get pilot estimates for β_0 and β_1 - Use these to get pilot residuals - Use pilot residuals to get pilot weights - Use weights in WLS to get updated estimates of β_0 and β_1 - Repeat steps 2-4 til convergence of weights
$Y = \beta_0^* + \beta_1^* \left(\frac{x_i - \bar{x}}{s_X} \right) + \dots + \beta_k^* \left(\frac{x_i - \bar{x}}{s_X} \right)^k + \epsilon$	$Y = \beta_0 + \beta_1 x + \dots + \beta_k x^k + \epsilon$ Problems: - Powers of x tend to be highly correlated - If k is large, then the magnitudes of the powers tend to vary over a very wide range - Overfitting
$Y_i = \beta_0 + \beta_1 x_{1i} + \beta_2 x_{2i} + \dots + \beta_p x_{pi} + \epsilon_i$ where $i = 1, \dots, n$ represents the observations and $\epsilon_i \stackrel{iid}{\sim} N(0, \sigma^2)$ $p + 2$ unknown parameters: $p + 1$ coefficients and σ^2	Occurs between 2 predictor variables when the effect on the response variable of one predictor depends on the value of the other predictor
(Same as SLR Model Assumptions) - Linear relationship between Y and x_j, $j = 1, \dots, p$ - Observations are independent - Constant variance - Normally distributed errors	$Y_i \stackrel{iid}{\sim} N(\beta_0 + \beta_1 x_{1i} + \dots + \beta_p x_{pi}, \sigma^2)$
$\hat{\beta} = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{Y}$	$\mathbf{Y} = \mathbf{X}\beta + \epsilon$ $\mathbf{Y}$ is an $n \times 1$ vector of observable r.v.s $\mathbf{X}$ is an $n \times (p + 1)$ matrix of known constants β is a $(p + 1) \times 1$ vector of unknown regression parameters ϵ is an $n \times 1$ vector of unobservable random errors such that $\epsilon \sim N_n(\mathbf{0}, \sigma^2 \mathbf{I})$

Estimate of σ^2 for MLR	Pivot for β_k
Hypothesis Test for Coefficients in MLR	ANOVA table for MLR
Global ANOVA Test for MLR	Adjusted R^2
Standardizing Regression Coefficients	Nominal Variable
Parallel Regression Lines (Using gender and height example)	Equal Intercept/Different Slope Regression Lines (Using gender and height example)

$\frac{\hat{\beta}_k - \beta_k}{\text{SE}\left(\hat{\beta}_k\right)} \sim t_{(n-p-1)}$	$\hat{\sigma}^2 = \frac{\text{RSS}}{n - (p + 1)} = \frac{\sum_{i=1}^n (y_i - \hat{y}_i)^2}{n - p - 1} = \text{MSE}$																				
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Source	df	SS	MS	F																	
Reg	p	SSR	$MSR = \frac{SSR}{p}$	$\frac{MSR}{MSE}$																	
Error	n - p - 1	SSE	$MSE = \frac{SSE}{n - p - 1}$																		
Total	n - 1	SST																			
Penalizes adding additional predictors to the model. Define to be $R^2_{\text{adj}} = 1 - \frac{\text{SSE}/(n - p - 1)}{\text{SST}/(n - 1)}$	$H_0 : \beta_1 = \beta_2 = \dots = \beta_p = 0 \text{ versus } H_a : \exists \beta_i \neq 0$ Useful to check if any of the predictors are helpful in explaining the response																				
A categorical variable with no natural ordering	Useful to judge the importance of different predictors by allowing for comparison of estimated regression coefficients $y_i^* = \frac{y_i - \bar{y}}{s_y} \quad \text{and} \quad x_{ij}^* = \frac{x_{ij} - \bar{x}_j}{s_{x_j}}$																				
$Y = \beta_0 + \beta_1 \cdot \text{height} + \beta_3 \cdot \text{gender} \cdot \text{height} + \epsilon$ where gender = 1 for females and 0 for males $Y = \begin{cases} \beta_0 + \beta_1 \cdot \text{height} + \epsilon, & \text{males}(\text{gender} = 0) \\ \beta_0 + (\beta_1 + \beta_3) \cdot \text{height} + \epsilon, & \text{females}(\text{gender} = 1) \end{cases}$ $H_0 : \beta_3 = 0$ tests for different slopes between the two groups	$Y = \beta_0 + \beta_1 \cdot \text{height} + \beta_2 \cdot \text{gender} + \epsilon$ where gender = 1 for females and 0 for males $Y = \begin{cases} \beta_0 + \beta_1 \cdot \text{height} + \epsilon, & \text{males}(\text{gender} = 0) \\ (\beta_0 + \beta_2) + \beta_1 \cdot \text{height} + \epsilon, & \text{females}(\text{gender} = 1) \end{cases}$ $H_0 : \beta_2 = 0$ tests for different intercepts between the two groups																				

Unrelated Regression Lines (Using gender and height example)	Hat Matrix
Vector of Fitted Values ($\hat{Y}$)	Residuals ($\hat{e}$)
$SE(e_i)$ (for MLR)	i th Standardized Residual (for MLR)
Checking MLR Assumption	Multivariate Cook's Distance
Partial R^2	Added Variable Plots

• $H = X (X^\top X)^{-1} X^\top$ • symmetric and idempotent	$Y = \beta_0 + \beta_1 \cdot \text{ht} + \beta_2 \cdot \text{gnr} + \beta_3 \cdot \text{gnr} \cdot \text{ht} + \epsilon$ where gender = 1 for females and 0 for males $Y = \begin{cases} \beta_0 + \beta_1 \cdot \text{ht} + \epsilon, & \text{males (gender = 0)} \\ (\beta_0 + \beta_2) + (\beta_1 + \beta_3) \cdot \text{ht} + \epsilon, & \text{females (gender = 1)} \end{cases}$ $H_0 : \beta_2 = 0$ tests for different intercepts between the two groups. $H_0 : \beta_3 = 0$ tests for different slopes between the two groups.
$\hat{e} = Y - \hat{Y} = Y - HY = (I - H) Y$	$\hat{Y} = \begin{bmatrix} \hat{y}_1 \\ \vdots \\ \hat{y}_n \end{bmatrix} = \begin{bmatrix} \mathbf{x}_1^\top \hat{\beta} \\ \vdots \\ \mathbf{x}_n^\top \hat{\beta} \end{bmatrix} = X \hat{\beta} = X (X^\top X)^{-1} X^\top Y$
$r_i = \frac{e_i}{\sqrt{\hat{\sigma}^2 (1 - h_{ii})}}$	$\sqrt{\hat{\sigma}^2 (1 - h_{ii})}$
$D_i = \frac{(\hat{\beta}_{(i)} - \hat{\beta})^\top X^\top X (\hat{\beta}_{(i)} - \hat{\beta})}{(p + 1) \cdot \hat{\sigma}^2} \quad \text{for } i = 1, \dots, n$	1. Check linearity by plotting standardized residuals against each individual predictor variable. 2. Check for constant variance by plotting the standardized residuals against the fitted values. 3. Check for heteroscedasticity by plotting the square root of the absolute standardized residuals against the fitted values. 4. Check for normality of errors by making a normal QQ plot of the standardized residuals.
These are the visual representation of partial R^2 in which the effect of each predictor on the response can be visually assessed after having adjusted for the effects of the other predictors	• Measures the degree of association between two random variables with the effect of a set of controlling variables removed • i.e. the effect of predictor 1 on the response given that the other predictors are fixed

Creating Added Variable Plots	Multicollinearity
Consequences of Multicollinearity in MLR model	3 ways to investigate multicollinearity
Four Selection Criteria	AIC
AIC_c	BIC
Backwards Elimination	Forward Selection

1. Present in MLR when the columns of $\mathbf{X}$ are exactly or approximately linearly dependent 2. Means that the predictors are linearly related 3. Problematic because it means that $\mathbf{X}^\top \mathbf{X}$ is no longer invertible	Let Y be the response with potential predictors $\mathbf{x} = (x_1, \dots, x_p)$. To assess the effect of x_1 on Y with the effect of the other variables, $\mathbf{x}_{(-1)} = (x_2, \dots, x_p)$ removed. 1. Regress x_1 on $\mathbf{x}_{(-1)}$ to obtain residuals, $\hat{e}_{x_1}$ 2. Regress Y on $\mathbf{x}_{(-1)}$ to obtain residuals, $\hat{e}_Y$ 3. Regress $\hat{e}_Y$ on $\hat{e}_{x_1}$, and get the R^2 for this model. This is the partial R^2, or the proportion of Y that x_1 explains with the effect of the other predictors removed
1. Correlation matrix of the predictors, $\mathbf{R}$ 2. The determinant of $\mathbf{R}$ 3. Variance inflation factors (VIF): VIF_j greater than 5 are generally unacceptable	1. The estimate $\hat{\beta}$ is subject to numerical errors and is unreliable. Small changes in the data may cause large changes in $\hat{\beta}$, and the sign of the coefficients may even be reversed. 2. The estimated coefficients will have very large standard errors, making them individually non-significant even if the overall F test is significant.
• Variable selection criterion • Based on the fitted log-likelihood • Tends to overfit models • Small values are considered better	• Adjusted R^2 • Akaike's Information Criterion (AIC) • Bias-Corrected AIC (AIC_c) • Bayesian Information Criterion (BIC)
• Based on posterior probability • Useful for models that make predictions at values of the explanatory variables that were not observed in the data (still within range of values) • Small values are considered better	• Variable selection criterion • Bias-corrected version of AIC • Converges to AIC as the sample size increases • Small values are considered better
• Begins with an initial model that only includes an intercept and add variables to the model one at a time • Stops when all variables have been added to the model or no information criterion decrease can be obtained by adding a variable • Useful for correlated data such as spatial or time series data	• Begin with the full model and sequentially eliminate the least important variable from the model • Stops when all variables have been deleted from the model or no information criterion decrease can be obtained by removing a variable • Starting with the full model is considered an advantage of this method

Training and Testing Sets: Large Dataset with Independent Observations	Training and Testing Sets: Large Dataset with Independent Observations (Leave m out cross validation)
Training and Testing Sets: Large Dataset with Dependent Observations	Different Types of Data
LASSO (Least Absolute Shrinkage and Selection Operator)	Ridge Regression
How λ affects coefficient estimates in LASSO and Ridge Regression	Oracle Properties
Adaptive LASSO	Bootstrap Sample Steps

1. Split data into many small subsets, each of size m 2. Remove m observations from the data 3. Train model on remaining $n - m$ observations 4. Use fitted model to predict withheld observations 5. Repeat for all subsets 6. Repeat for different tuning parameters in model or different types of models	1. Randomly assign 60% of the data to training and 40% to testing 2. Fit the model to the training data 3. Use the fitted model to predict the values in the testing data 4. Repeat steps 3-4 as needed for different tuning parameters in the model or different types of models
• Predictive: used to predict the response variable for a chosen set of predictor variables • Theoretical: used to assess the validity of a theoretical relationship between the response and predictors • Control: used to control the response by manipulating the predictors • Inferential: used to explore the strength of the relationship between response and predictors • Data Summary: used to summarize a large set of data with a single equation	1. Assign first 60% of data to training and the second 40% to testing 2. Fit several candidate models to the training data 3. Evaluate the “good” models. Check for any violations of assumptions or need for transformations 4. Select the final model and compare the competing models by cross-validating them against the test data
• Like LASSO, but cannot send coefficients to 0 only close to 0 • Better to use if the predictions are strongly correlated to each other or if the number of observations is less than the number of predictors $\min \sum_{i=1}^n (y_i - (\beta_0 + \beta_1 x_{1i} + \cdots + \beta_p x_{pi}))^2 + \lambda \sum_{j=1}^p (\beta_j)^2$	• Allows some bias into the estimates of the regression coefficients to reduce the variance of the model fit • Uses all of the variables, but penalizes large coefficient values and drives those that are small to 0 • A LASSO coefficient of 0 implies that we may eliminate the variable $\min \sum_{i=1}^n (y_i - (\beta_0 + \beta_1 x_{1i} + \cdots + \beta_p x_{pi}))^2 + \lambda \sum_{j=1}^p \beta_j $
Having oracle properties means that on average, the true subset of variables is chosen for the model	• Higher values of λ are more constrained, pushing the coefficient values closer to 0 • Lower values of λ are less constrained, pushing the coefficients to be similar to the least squares estimates • When λ equal 0, then the least squares estimates of the coefficients are obtained
1. Select n values from a dataset of size n with replacement. Then this bootstrap sample is $y_b^* = (Y_{b1}^*, \dots, Y_{bn}^*), b = 1, \dots, B$ 2. For each bootstrap sample, calculate the statistic of interest, denoted $\hat{\theta}_b^*$ 3. Use the distribution of the $\hat{\theta}_b^*$'s to estimate the properties of the sampling distribution of $\hat{\theta}$ 4. The bootstrap SE of $\hat{\theta}$ is $SE^*(\hat{\theta}^*) = \sqrt{\frac{\sum_{b=1}^B (\hat{\theta}_b^* - \bar{\theta}^*)^2}{B-1}}$	• Uses a pilot estimate (typically LS estimates or ridge estimates) for the coefficients, and drives those that are initially close to 0 faster • Has oracle properties $\min \sum_{i=1}^n (y_i - (\beta_0 + \beta_1 x_{1i} + \cdots + \beta_p x_{pi}))^2 + \lambda \sum_{j=1}^p \omega_j \beta_j $ where ω_j are weights based on the initial pilot estimates of the coefficients: $\omega_j = \left \hat{\beta}_j^{\text{initial}} \right ^{-\gamma}, \gamma = .5, 1, \text{ or } 2$

Normal Theory Intervals	Percentile Intervals
Bias-Corrected Intervals	Bootstrap Sample from a Regression Model (Treating the Predictors as Random)
Bootstrap Sample from a Regression Model (Treating the Predictors as Fixed)	Kernel Density Estimator (KDE)
Nadaraya-Watson Estimator	Cross-Validation
Nearest Neighbor Bandwidth	Lowess vs Loess Smoothing

$\left(\hat{\theta}_{(\text{lower})}^*, \hat{\theta}_{(\text{upper})}^*\right)$ where lower = $B \cdot \alpha/2$, and upper = $B \cdot (1 - \alpha/2)$	$\hat{\theta} \pm z_{\alpha/2} \cdot \text{SE}^* \left(\hat{\theta}^*\right)$ • This works well if the bootstrap sampling distribution is approximately normal • However, there is no big advantage to using the bootstrap standard error over an interval based on the t-distribution if the sampling distribution of $\hat{\theta}$ is approximately normal
1. Let $\mathbf{z}'_i = [Y_i, X_{i1}, \dots, X_{ip}]$, $i = 1, \dots, n$ 2. Resample from $\mathbf{z}'_i$ with replacement n times 3. Fit the linear model on the resampled data, saving the estimate of interest 4. Repeat steps 2 and 3, B times	Corrects for bias in the raw percentile interval $\left(\hat{\theta}_{(\text{lower})}^*, \hat{\theta}_{(\text{upper})}^*\right)$ where lower = $B \cdot A_1$, and upper = $B \cdot A_2$ NOTE: Look at page 146 in notes for equations for A_1 and A_2
Let $X_1, \dots, X_n$ denote a sample of size n from a random variable with density function $f(\cdot)$. The KDE of $f(\cdot)$ at the point x is given by $\hat{f}_h(x) = \frac{1}{nh} \sum_{i=1}^n K\left(\frac{x - X_i}{h}\right)$ where the kernel $K(\cdot)$ satisfies $\int K(x) dx = 1$, and the smoothing parameter, h, is known as the bandwidth. Typically we use the Gaussian kernel $K(y) = (2\pi)^{-1/2} \exp(-y^2/2)$	1. Estimate the regression coefficients, $\{\beta_0, \beta_1, \dots, \beta_p\}$, for the original sample. Calculate the fitted values, $\hat{Y}_i$, and the residuals, $\hat{e}_i$ 2. Select bootstrap samples of the residuals, $(\hat{e}_{b1}^*, \dots, \hat{e}_{bn}^*)$, and use these to generate bootstrapped Y values, $\mathbf{y}_b^* = (Y_{b1}^*, \dots, Y_{bn}^*)^\top$, where $Y_{bi}^* = \hat{Y}_i + \hat{e}_{bi}^*$ 3. Regress the new $\mathbf{y}_b^*$ on the observed X values, and save the bootstrapped estimate of interest 4. Repeat steps 2 and 3, B times
A basic way to choose h is via cross-validation where h is chosen such that the following criteria is minimized: $\text{CV}(h) = \frac{1}{n} \sum_{i=1}^n (Y_i - \hat{m}_i(x_i; h))^2,$ where $\hat{m}_i(x_i; h)$ is an estimator of Y_i that is computed without using the ith data values, (x_i, Y_i)	$b_0 = \frac{\sum_{i=1}^n y_i K\left(\frac{x_i - x}{h}\right)}{\sum_{i=1}^n K\left(\frac{x_i - x}{h}\right)}$
• Lowess uses local linear regression estimators with nearest neighbor bandwidth and the tricube kernel function that also uses a robustness step to downweight large residuals • Loess is commonly used in practice. It is like lowess but does not use the robustness step of the lowess estimator	The bandwidth $h(x)$ is defined to be the kth smallest value of $d_i(x) = x - x_i , i = 1, \dots, n$ The choice of k is based on what is called the span α, namely, $k = \lfloor n\alpha \rfloor$. Does not perform well with bimodal or multimodal distributions

Treatment	One-Way ANOVA Main Effects Model
One-Way ANOVA Treatment Effects Model	One-Way ANOVA Assumptions
Cell Mean Model Hypothesis Test	Treatment Effects Hypothesis Test
ANOVA SST	ANOVA SSA
ANOVA SSE	NOTE:

$Y_{ij} = \mu_i + \epsilon_{ij}, \quad i = 1, 2, \dots, a, \quad j = 1, 2, \dots, n_i$ where $\epsilon_{ij} \sim N(0, \sigma^2)$ - Unknown parameters: $\mu_1, \dots, \mu_a$ and σ^2 - Residuals: $e_{ij} = y_{ij} - \hat{y}_{ij}$ where $\hat{y}_{ij} = \hat{\mu}_i$	- A particular combination of levels of factors - the number of treatments equals the number of levels of each factor multiplied together
ϵ_{ij} are normally distributed - Constant variance in each treatment (more crucial assumption)	$Y_{ij} = \mu + \tau_i + \epsilon_{ij}, \quad i = 1, \dots, a, \quad j = 1, \dots, n_i$ where $\epsilon_{ij} \sim N(0, \sigma^2)$ - Unknown parameters: μ, σ^2 and $\tau_1, \dots, \tau_a$ - Residuals: $e_i = y_{ij} - \hat{y}_{ij}$ where $\hat{y}_{ij} = \hat{\mu} + \hat{\tau}_i$ - NOTE: If $\tau_i = 0$, then the ith treatment has no effect on the overall mean
$H_0 : \tau_1 = \tau_2 = \dots = \tau_a = 0$ $H_a : \text{At least one } \tau_i \neq 0$	$H_0 : \mu_1 = \mu_2 = \dots = \mu_a$ $H_a : \text{At least one } \mu_i \text{ differs}$
Describes the differences between treatment means (sum of squares due to treatment A) $SSA = \sum_{i=1}^a \sum_{j=1}^{n_i} (\bar{y}_{i\cdot} - \bar{y}_{\cdot\cdot})^2$	Describes the total variability in the y_{ij} $SST = \sum_{i=1}^a \sum_{j=1}^{n_i} (y_{ij} - \bar{y}_{\cdot\cdot})^2$
Look at pages 163-164 in the notes to understand the ANOVA table (it was too large to put on a flashcard)	Describes the differences between individual measurements within each treatment group $SSE = \sum_{i=1}^a \sum_{j=1}^{n_i} (y_{ij} - \bar{y}_{i\cdot})^2$

Familywise Error Rate (FWE)	Three approaches to controlling FWE to be α
Contrast	Sampling Distribution of Sample Estimator for a Contrast
Three Methods for Examining Multiple Contrasts	Checking ANOVA Assumptions
Fixed Effects Model	Random Effects Model
Two-Way ANOVA Cell Means Model	Two-Way ANOVA Treatment Effects Model

• Fisher's LSD: Only test all pairwise hypotheses when the ANOVA F-test is rejected. Not conservative enough. • Bonferonni Method: IN this approach, the FWE rate is divided among the k tests, and each test is done at level α/k. Hyper-conservative approach. • Tukey Method: In a balanced one-way ANOVA, the Tukey method exactly controls the FWE to be α and is only slightly conservative for unbalanced designs. Generally the best approach.	FWE = P (Reject at least one true null hypothesis) We want to control the FWE to be α where the "family" refers to the collection of all $k = \binom{a}{2}$ pairwise null hypotheses
$\sum_{i=1}^a c_i \bar{Y}_i \sim N \left(\sum_{i=1}^a c_i \mu_i, \sigma^2 \sum_{i=1}^a \frac{c_i^2}{n_i} \right)$	• A linear function of the μ_i, denoted $\sum_{i=1}^a c_i \mu_i$, such that the coefficients sum to 0, i.e., $\sum_{i=1}^a c_i = 0$ • The sample estimator for a contrast is $\sum_{i=1}^a c_i \bar{Y}_i$
• Normality: Plot ordered standardized residuals against the corresponding normal quantiles or apply the Shapiro-Wilks Tests • Constant Variance: Plot the standardized residuals against the fitted values from the model. There should not be any non-linear relationship. Also apply Bartlett's or Levene's test which tests – $H_0 : \sigma_1^2 = \sigma_2^2 = \dots = \sigma_a^2$ – $H_a : \text{At least one variance is different}$	• Dunnett Method: Use when there is a control group or benchmark for comparison. In a balanced one-way ANOVA, if group 1 is the control, the contrasts of interest are $\mu_2 - \mu_1, \dots, \mu_a - \mu_1$ • Scheffe Methods: "All possible contrasts" • Hsu Method" Used to find the "best" treatment
• The only factors in the model are those that are random, i.e., the treatments for each factor are randomly chosen • This model is also called the "variance components model" because it expresses the variance of the response as the sum of the variance between the levels of the treatment factor and the error variance	The only factors in the model are those that are fixed, i.e., the treatments for each factor are the only ones of interest to the researcher
$Y_{ijk} = \mu + \tau_i + \beta_j + (\tau\beta)_{ij} + \epsilon_{ijk}, \quad \epsilon_{ijk} \sim N(0, \sigma^2)$ • Main Effects: $\tau_i = \mu_{.i} - \mu$ and $\beta_j = \mu_{.j} - \mu$ • Interaction Effect: $(\tau\beta)_{ij} = \mu_{ij} - \mu - \tau_i - \beta_j$	$Y_{ijk} = \mu_{ij} + \epsilon_{ijk}, \quad \epsilon_{ijk} \sim N(0, \sigma^2)$

Row Main Effect	Column Main Effect
Interaction Effect	Interaction Plot
NOTE:	

β_j is the difference between the mean response of the jth level of factor B (averaged over all levels of factor A) and the grand mean	τ_i is the difference between the mean response of the ith level of factor A (averaged over all levels of factor B) and the grand mean
• The cell sample means are plotted with levels of one factor on the x-axis and levels of the other factor represented by different line types • Parallel lines indicate that the difference in levels of one factor is the same no matter what the level of the other factor is	$(\tau\beta)_{ij}$ is the part of the cell mean, μ_{ij}, that is over and above the separate contributions from the grand mean, row main effect, and column main effect
	Look at pages 179-180 in the notes to understand the Two-Way ANOVA table (it was too large to put on a flashcard)